Ruiwen WANG | Curriculum Vitae

ML Systems · AI Infrastructure · Distributed Computing
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Scale & deployment: Evaluated BMPipe on 16,384 NPUs; helped automate parallelism tuning for TeleChat3-MoE.
Open source: Contributed to pipeline balancing in MindSpore and MindFormers. **Research:** 7 publications · 3 patent families.

Experience

Huawei Paris Research Center · Distributed & Parallel Computing Lab — Paris, France

Ph.D. Researcher (CIFRE) *In collaboration with Sorbonne Université & EURECOM* — 2023 - 2026

- Designed symbolic cost models and hybrid-parallel planners that jointly optimise parallelism, pipeline placement and recomputation under memory constraints; benchmarked against Megatron-LM.
- Integrated planners into MindSpore / MindFormers (open-source AI training stack; SAPP-PPB), enabling production use through configuration guidance, diagnostic traces and regression benchmarks across model families and cluster scales.

Research Engineer — 2022 - 2023

- Contributed to MindSpore's automatic-parallelisation framework, using symbolic cost analysis and recursive graph partitioning to generate operator-level sharding strategies for distributed DNN training.

Research Apprentice — 2021 - 2022

- Profiled and tuned large-scale LLM training and inference on production Ascend 910 NPU clusters, using execution-trace analysis to diagnose compute, memory and communication bottlenecks.

Deployments & Impact

- **Production training:** Helped automate parallelism tuning for China Telecom's TeleChat3-MoE, reducing tuning time from about a week to half a day while matching expert-tuned step times on 4,096 Ascend NPUs.
- **Scale:** Evaluations ran on up to 16,384 Ascend 910 NPUs. Across projects, models spanned dense, MoE and multimodal architectures; some had over 1T parameters. Families included DeepSeek, Llama, Mistral/Mixtral and Qwen.

Publications

First-author publications

- **Concord** - NeurIPS 2026. For iterative model changes, symbolic layer templates expose costs, dependencies and value signatures for safe evaluation reuse, accelerating re-planning while preserving full-search semantics. Joint optimisation of hybrid-parallel configurations, layer placement and recomputation reduces training step time.
- **PRISM** - SCA/HPC Asia 2026. Memory feasibility depends on parallelism, scheduling and recomputation. Symbolic memory and communication models evaluate these choices without per-plan profiling, enabling pre-execution peak-memory prediction.
- **BMPipe** - IEEE CLUSTER 2025. Pipeline stages waste compute time and leave memory unused. Joint symbolic modelling and ILP optimise placement, chunking and recomputation under memory limits, improving throughput at production scale.
- **H^2O** - Euro-Par 2025, Best Poster Award. Joint planning is combinatorial. A cost-guided hierarchy couples parallelism and micro-batching with pipeline placement and recomputation, improving throughput over automated and expert-tuned plans.
- **ManuMatic** - IFIP NPC 2025. Cost models can miss runtime overlap and new kernels. A few user-specified operator shardings steer recursive partitioning and graph-consistent completion, improving throughput over the automatic baseline.

Co-authored publications

MLLM Pipeline Bubble Modeling - SCA/HPC Asia 2026 (poster).
TeleChat3-MoE - arXiv 2025 (technical report).

Expertise & Technical Skills

Parallelism: DP / TP / PP / EP; pipeline scheduling; recomputation. **Planning:** cost modelling; ILP; graph partitioning.
Performance: profiling; trace analysis; bottleneck diagnosis. **Programming:** Python, C++. **Tools:** Linux, Git.
Frameworks: PyTorch, Megatron-LM, MindSpore, MindFormers. **Systems:** Ascend CANN, HCCL.
Languages: English, French, Mandarin, Cantonese.

Patents

Execution-plan generation - WO 2025/020165 A1 (PCT), Huawei; published 2025.
Pending publication (Huawei): Large-model load balancing; Multidimensional parallelism configuration.

Education

Ph.D. Computer Science (CIFRE) - Sorbonne Université — Supervisors: Raja Appuswamy, Chong Li — 2026 (defended)

M.Sc. Computer Science - Sorbonne Université — Advisors: Emmanuel Chailloux, Jacques Malenfant — 2019 - 2022

B.Sc. Computer Science - Université Paris-Saclay — Mentors: Isabelle Guyon, Viviane Pons, Zhengying Liu — 2016 - 2019

Awards & Recognition

Best Poster Award - Euro-Par · Letter of Appreciation - China Telecom (TeleChat3-MoE training) — 2025
Golden Prize - Huawei — 2024